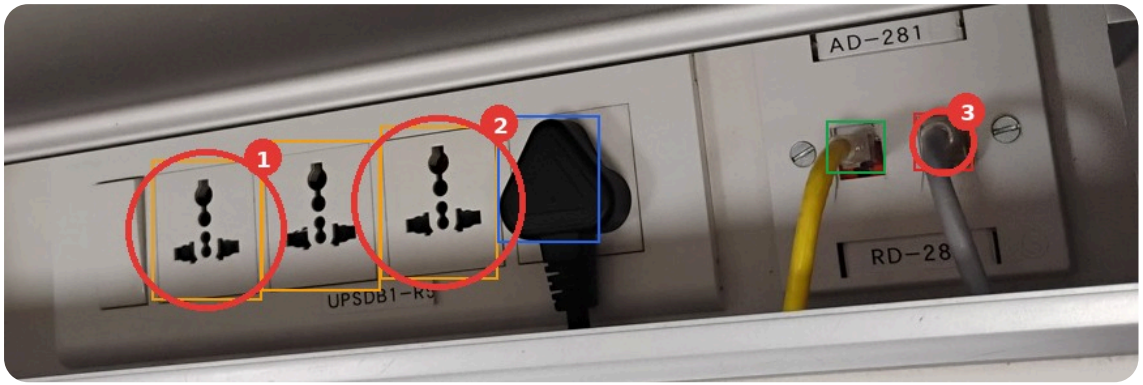
On-Device AI Install Checker — YOLO engine

Same app, same checklist — a second, much smaller AI engine: YOLO object detection + fixed rules.



- Model: **YOLO26n** (Ultralytics), trained on our 9 photos
- **5 MB** instead of 3.86 GB — downloads in seconds
- Finds **14 / 14** problems on the demo photos (Gemma: 12 / 14)
- One model, **iOS (Core ML)** and **Android (TFLite)**
- Fully offline · switch engines in the app's *About* screen

1 · How does the AI "see" a photo?

The AI does one job: it finds every socket and port, and names what is in it. A short checklist then decides pass or fail.



① Photo
the technician takes a picture



② AI finds & names
"empty, empty, empty, plug, LAN, extra cable"

FAIL · 3 issues
- × socket 1: adapter missing
- × socket 3: plug missing
- × grey cable plugged in
- ✓ LAN cable OK

③ Checklist decides
compare with the standard setup

- This kind of AI is called **object detection**. It is the same idea a car uses to spot pedestrians, just trained on our 6 things.
- To teach it, we show it photos where a person has already drawn and named every box. The next 5 slides are those steps.

2 · Training step 1 — Collect photos

1 · Collect photos | 2 · Draw boxes | 3 · Make variations | 4 · Practice (train) | 5 · Test & package

Take real photos of the desk box, both correct and wrong installs.



- **9 photos** for the demo: 4 correct, 5 with mistakes (14 mistakes in total).
- Every kind of mistake needs examples: a missing adapter, a missing plug, a wrong cable.
- Photos are straightened and resized the same way the app will do it on the phone.
- **For production:** 30–50 photos per situation, from different rooms, lighting and angles.

3 · Training step 2 — Draw boxes and name them

1 · Collect photos | **2 · Draw boxes** | 3 · Make variations | 4 · Practice (train) | 5 · Test & package

A person draws a box around every socket and port and picks its name. This is the "answer key" the AI learns from.



Name	Meaning
adapter	power adapter plugged in
plug	power cord plugged in
empty	empty power socket
LAN	yellow LAN cable
extra	any other cable
port	empty data port

- **53 boxes** on 9 photos. Tools like Label Studio or Roboflow make this point-and-click.
- We label *what we see*, not "right" or "wrong". If the standard setup changes, the labels stay the same.

4 · Training step 3 — Turn 9 photos into thousands

1 · Collect photos | 2 · Draw boxes | **3 · Make variations** | 4 · Practice (train) | 5 · Test & package

Each time the AI practices, the photos are shuffled, tilted, zoomed and stitched together, so it never sees the exact same picture twice.

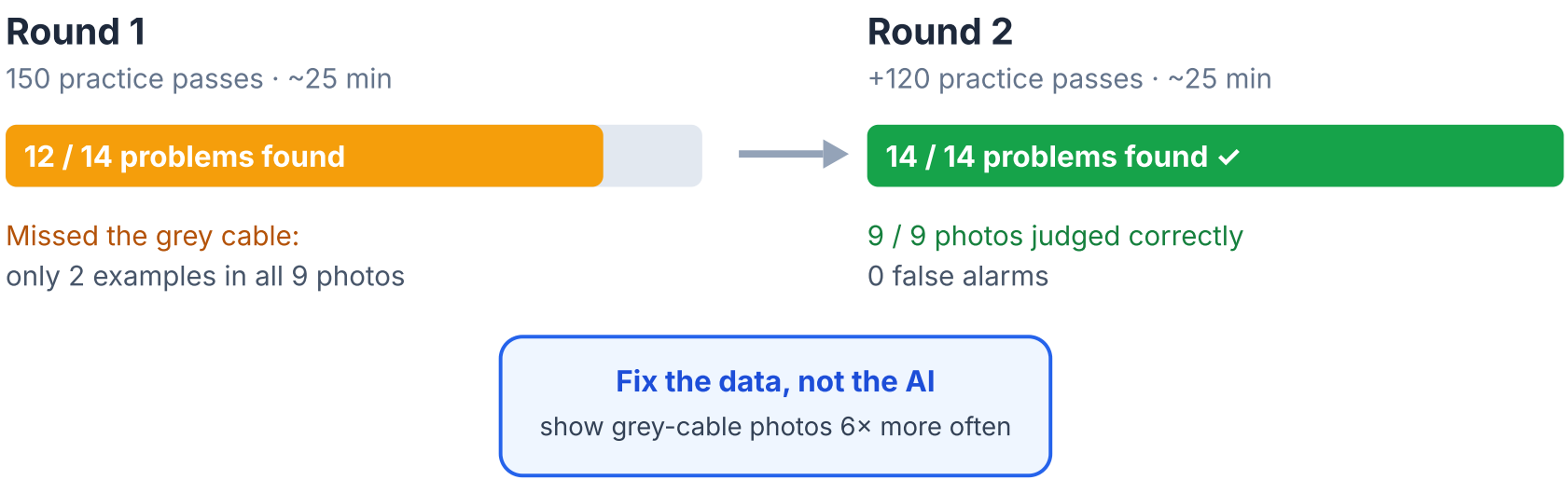


- Like showing a new employee the same desk from different angles and in different light.
- The boxes move with the picture, so the answer key stays correct.
- We keep the **colors** almost unchanged, because yellow vs. grey cable is exactly what the AI must notice.
- Rare mistakes (the grey cable) are shown more often so the AI does not ignore them.

5 · Training step 4 — Practice, check, repeat

1 · Collect photos | 2 · Draw boxes | 3 · Make variations | **4 · Practice (train)** | 5 · Test & package

The AI guesses the boxes, compares with the answer key, and corrects itself, hundreds of times.

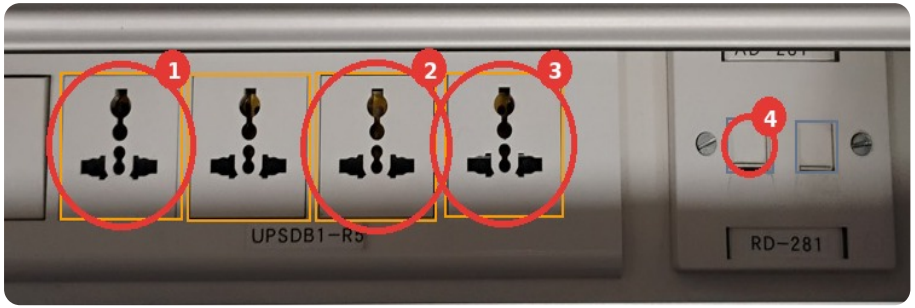


- We start from a model that already knows how to find everyday objects, and only teach it our 6 new names. That is why 9 photos are enough for a demo.
- Runs on a normal laptop: about **50 minutes** in total, no cloud and no special hardware.

6 · Training step 5 — Test and package for the phone

1 · Collect photos | 2 · Draw boxes | 3 · Make variations | 4 · Practice (train) | **5 · Test & package**

Check the AI + checklist against the 14 known mistakes, then shrink it into one small file the phone can run.



wrong install → red circles

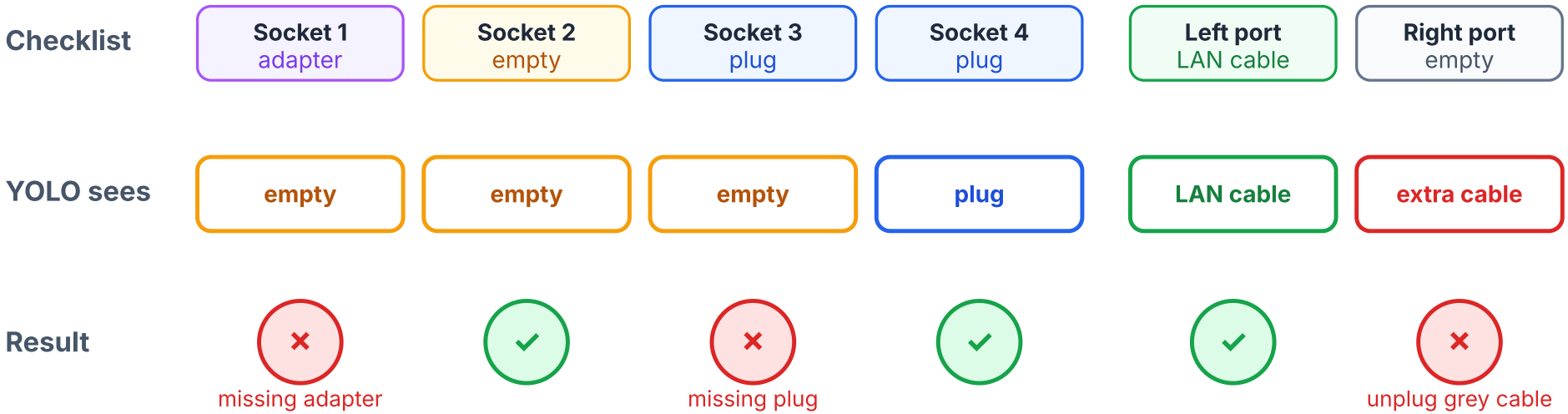


correct install → PASS

Test result	Packaged file
14 / 14 mistakes found · **9 / 9** photos judged correctly · 0 false alarms	**iPhone:** Core ML, 4.9 MB · **Android:** TFLite, 9.9 MB · uploaded to GitHub, the app downloads it in seconds

7 · From boxes to problems: the checklist rules

YOLO only says *what is in each slot*. A few lines of code compare that with the standard setup.



- Sockets are sorted **left → right** and must be: adapter · empty · plug · plug. The data ports need **one yellow LAN cable** and **no other cable**.
- Every mismatch becomes one red circle around that slot, plus a fixed fix-it sentence, e.g. *"Missing power plug. Plug a power cord into the third socket."*
- The rules live in `InstallRules.swift` (app) and `yolo/rules.py` (training repo). Changing the checklist means changing the rules, not retraining.